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Sensitivity of circulation, thermal structure, and transient features to surface forcing in a Lake Erie FVCOM model

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ABSTRACT

Lake Erie, the most urbanized and biologically productive of the Great Lakes, currently has multiple operational or semi-operational systems which are used to produce model-generated nowcasts and forecast guidance for water temperatures, currents, water levels, lake ice, harmful algal blooms and hypoxic bottom waters throughout the lake. It is likely that a major source of error in these products is error in the surface wind forcing, given Lake Erie's relative shallowness and the sparsity of over-lake wind observations. This work investigates the sensitivity of a 2016 version of the FVCOM model used for these systems to the surface forcing used to drive the model. Two different forcing datasets that have been used operationally are used to drive the model for a year, resulting in substantial differences in model circulation and thermocline shape and depth, as well as in transient phenomena such as seiches and near-inertial oscillations. These results highlight some common long-term and short-term forecasting challenges for lakes with wind-driven circulation, and point to the value of using data assimilation (DA) to adjust the forecast initialization to more closely match recent observations, a process not yet used in Great Lakes operational forecast systems. The use of an ensemble-based DA system for Lake Erie will be explored in a forthcoming work.

1. Introduction

The Great Lakes Operational Forecast System (GLOFS, Schwab and Bedford 1994, Anderson et al. 2018) generates model-based nowcasts and forecast guidance for temperature, current velocity, and water level throughout the lakes, as well as lake ice concentration, thickness and velocity where applicable. It currently uses four constituent free-running hydrodynamic models of the lakes based on the Finite-Volume Community Ocean Model (FVCOM; Chen et al. 2003, 2011), coupled to the sea ice model UG-CICE (Gao et al., 2011), with atmospheric boundary conditions at the free surface coming from two sources: For the nowcast (from 24 h previous to the current time), the National Oceanic and Atmospheric Administration (NOAA) High-Resolution Rapid Refresh (HRRR) model (Dowell et al., 2022) hour-1 forecast is used, and for the 5-day forecast (initialized from the nowcast), a National Digital Forecast Database (NDFD) forecast is used (personal communication with Todd Ehret, NOAA-COOPS, 2025). A major upgrade to GLOFS came in 2016, when the old Princeton Ocean Model (POM) models were replaced by the FVCOM models, which use finite-volume methods on an unstructured grid, allowing for high resolution where needed, such as in shallow basins or near complex coastline topography. These models are now also used for seasonally-relevant forecast systems such as the Lake Erie Harmful Algal Bloom (HAB) Forecast (Rowe et al., 2016), and the Lake Erie Experimental Hypoxia Forecast (Rowe et al., 2019). Since 2016, GLOFS has continued to improve as newer versions of its constituent models and HRRR have been released, with FVCOM version 4.4.7 and HRRR version 4.1 being used at the time of writing.

However good the GLOFS models may be, errors are inevitable in any free-running model forecast, due to errors in both the model itself (e.g. simplifying assumptions, discretization of physical laws, sub-grid-scale processes) and in the initial and boundary conditions used by the model. The relative importance of these different sources of forecast error depends on the system in question. For example, in the global atmosphere, errors in the initial state are quite important as instabilities in the system can make these errors grow relatively quickly. In numerical weather prediction, this problem has long been addressed via data assimilation (DA), i.e. regular adjustment and re-initialization of the model state for the forecast to more closely match

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recent observations. At present, GLOFS does not use DA to re-initialize the model state for the forecast, so that forecast errors may potentially grow unchecked, especially in the subsurface where the effects of the changing surface fluxes may be slow to penetrate.

Lake Erie, the most urbanized and biologically productive of the Great Lakes, is also the shallowest by far, so errors in the surface boundary conditions (the wind forcing in particular) are likely to be a major driver of forecast error, compared to the other Great Lakes, in which density-driven currents play a larger role (Beletsky et al., 1999). After the switch to FVCOM models, hindcast and nowcast/forecast skill assessment reports for the Lake Erie Operational Forecast System (LEOFS, one component of GLOFS) showed significant deviations from observations of temperatures and water levels at some locations, which were hypothesized to result primarily from errors in the wind forcing (Kelley et al., 2018; Xu et al., 2018). Using this same FVCOM model to study hypoxia in the central basin, Rowe et al. (2019) showed the thermocline depth and shape there to be highly sensitive to the choice of wind forcing dataset used to drive the model, with mixed conclusions about which dataset led to a closer match to observed profiles of temperature and dissolved oxygen. This is consistent with other studies which have shown the thermocline shape (i.e. bowl or dome) to be sensitive to whether the wind stress curl is cyclonic or anticylconic, in both observations and a POM-based model (Beletsky et al., 2012, 2013). Due to these results, as well as Lake Erie's persistent hazards related to weather (lake-effect snow, seiche-induced flooding) and environmental stresses (HABs, hypoxia, invasive species, plastic pollution), recent years have seen a growing interest in using DA to improve LEOFS (Ye et al., 2020). Continuing this effort, a forthcoming paper by the authors will present and evaluate an ensemble-based DA method to improve LEOFS nowcasts.

This work paves the way for that one by providing a more complete accounting of the nature and magnitude of the model state errors that can result from wind forcing error with a free-running FVCOM model, during the summer months when HABs and hypoxic bottom waters tend to occur. Rather than validating the model against observations, this work builds on Rowe et al. (2019) by examining in greater depth the effects of the two surface forcing datasets used therein on a 2016 version of the LEOFS model. First, a set of four-month sensitivity experiments is used to assess the response times of the model temperature, velocity and free surface height to the surface forcing, the relative impacts of each forcing variable, and the relative impact of initialization versus forcing differences. Then, the two forcing datasets are compared through their mean fields and their empirical orthogonal function (EOF) decomposition (Jolliffe, 2011). Year-long free model runs are performed using each of the two datasets (to include several months of model spin-up encompassing the spring turnover and onset of stratification), and their differing effects on the model circulation and thermal structure during June-July-August are compared, in addition to their effects on two well-known transient phenomena which oscillate on sub-24-hour time scales, namely seiches and near-inertial oscillations (Boyce and Chiocchio, 1987; Saylor and Miller, 1987; Rao et al., 2008). This analysis, while of interest in its own right, also lays the groundwork for the forthcoming DA paper, in which a series of observing system simulation experiments (OSSEs) is conducted using one field as the "true" wind field, and the other as the "erroneous" wind field. In that work, the ensemble-based DA system attempts to correct the forecast errors largely caused by the erroneous winds, using simulated observations of the run using the true winds (the "nature run"), in combination with a forcing ensemble based on the EOF decomposition of both wind fields.

This work is organized as follows. In Section 2, the model, surface forcing datasets, and experimental setup are introduced. In Section 3, the results of initial sensitivity experiments are presented, the two forcing datasets are examined in detail, and the results of the two year-long free model runs forced by those datasets are compared through both their mean fields and their transient phenomena, using EOFs and spectral analysis of time series at multiple locations. Finally, Section 4 summarizes the main findings and their implications for forecasting, and points to the importance of augmenting the forecast model with a DA system, such as the one presented in the authors' forthcoming paper.

2. Methods

2.1. Forecast model

The Lake Erie FVCOM model used for this study was provided by NOAA's Great Lakes Environmental Research Laboratory (GLERL). The FVCOM version (3.2.0) and configuration are those used by LEOFS in 2016, prior to sea ice coupling. FVCOM uses finite-volume methods (Eymard et al., 2000) to solve the integral form of the 3D primitive equations of fluid flow (momentum, continuity, temperature, salinity, density) on an irregular horizontal grid with a terrain-following vertical coordinate and time-varying free surface. The prognostic model variables are temperature T, salinity S, horizontal velocity $\mathbf{v} = (u, v)$, and free surface height ζ, plus two turbulence variables required by the Mellor-Yamada 2.5 turbulence closure scheme (Mellor and Yamada, 1982), namely q^2 (the turbulent kinetic energy) and q^2l (its product with the turbulent macroscale). The model uses 20 vertical σ-layers of equal thickness, with horizontal resolution ranging from about 0.24 to 3.50 km (Fig. 1). The lateral open boundaries consist of five inflow nodes at the Detroit River, and two outflow nodes at the Niagara River. The model bathymetry reflects Lake Erie's usual division into three basins: a small, shallow western basin, a large, flat central basin, and a deeper, bowl-shaped eastern basin.

At the free surface, the model is forced by 2D wind velocity (v_w), air temperature (T_a), dew point temperature (T_d), and cloud fraction (C). Water levels are prescribed at both lateral open boundaries, and temperature follows a no-gradient condition at the outflow boundary to avoid heat transfer, while at the inflow boundary, this condition is compromised by nudging the boundary value toward a prescribed value in the forcing file. Salinity is set to zero throughout the initial model state and lateral boundary conditions. The model uses FVCOM's default mode-split time-stepping, consisting of an external mode that uses the vertically-averaged momentum and continuity equations to update the two-dimensional variables (ζ and the vertically-averaged velocities $\overline{u}$, $\overline{v}$), and an internal mode that uses the full momentum, continuity and tracer equations to update the three-dimensional variables. The time step is 10 s for both modes. The code is parallelized by horizontally partitioning the model domain into eight subdomains using the graph-partitioning software library METIS (Karypis and Kumar, 1997) and running one process per subdomain, with information exchanged at the interprocessor boundaries after each time step (Cowles, 2008).

2.2. Surface forcing datasets

Because Lake Erie's circulation is known to be highly wind-driven, this work compares the results from using two different surface forcing datasets to drive the model. Both forcing datasets, as well as the initial conditions described in Section 2.3 were supplied by NOAA GLERL (personal communication with Eric Anderson, 2016). Both datasets consist of hourly $\mathbf{v}_w$, T_a, T_d and C for all of 2017, and they coincide with those used in Rowe et al. (2019), where their differing effects are discussed in the context of hypoxia in the central basin. The first dataset is based on output from HRRR (Dowell et al., 2022), a cloud-resolving, convection-allowing model used to issue hourly forecasts for the continental United States by NOAA's National Centers for Environmental Prediction (NCEP). HRRR version 2 (operational in 2017) used a 3-km grid with boundary conditions from the coarser (13-km) Rapid Refresh model. Every hour, it assimilated high-frequency (15-min) three-dimensional radar reflectivity observations, in addition to

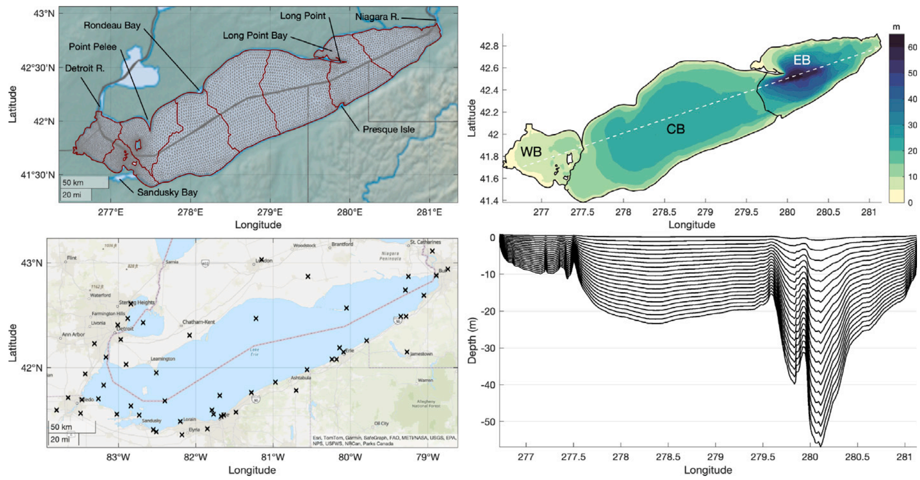


Fig. 1. Clockwise from top left: Model grid with domain partition in red; model bathymetry showing western basin (WB), central basin (CB) and eastern basin (EB); vertical transect showing model layers (transect line shown above); stations/platforms used for interpolated surface meteorology (as in Rowe et al. 2019).

conventional observations from various sources including aircraft, rawinsondes, surface stations, ships, buoys, profilers, and satellite winds. For this work, the HRRR output is then interpolated to the similar-resolution FVCOM model grid, using not only over-lake but also over-land HRRR values (whose inclusion was found to have negligible impact on the results of Rowe et al. 2019).

The second dataset is based not on model output, but rather on meteorological observations from 9 data buoys and 45 land-based stations, mostly lying near the US shoreline (Fig. 1). NOAA GLERL prepared these observations using processes detailed in Beletsky et al. (2003) and Schwab and Beletsky (1998). Briefly, wind speeds were adjusted from the observation height to a uniform height based on an assumed height profile, interpolated in space and time to the model grid using a natural neighbor method, and land-derived speeds were inflated to over-lake values based on empirical relationships. This dataset is hereafter referred to as 'Interp' due to the major role played by interpolation in filling in gaps between observations. This type of dataset was used to force the previous-generation (pre-2016) LEOFS based on the Princeton Ocean Model (POM), and has been used more recently for the FVCOM-based Lake Erie Experimental Hypoxia Forecast. Unlike for the HRRR data, this method does not allow for the representation of small-scale weather phenomena passing over the lake, due to the relatively sparse nature of the interpolation points (see Section 3.2 for a more detailed analysis of the differences between the two forcing datasets). (After completion of experiments, the HRRR dataset was found to coincide with the Interp dataset for two intervals totaling 8 h during Jun-Jul-Aug, because of a lapse in operational availability resulting in gap-filling, but such a short time is unlikely to have an appreciable effect on the analysis presented here.)

2.3. Approach and experimental setup

A list of experiments performed for this work is shown in Tables 1 and 2. The main experiments are two free model runs for all of 2017, one using the HRRR forcing, and the other using the interpolated ("Interp") forcing. The "HL" run (for "HRRR Long") is examined in Section 3.3, while the "IL" run ("Interp Long") is presented for comparison in Section 3.4. Both runs use spun-up initial conditions (without ice) for January 1st 2017, provided by NOAA GLERL.

Prior to in-depth analysis of the long runs, however, the results of several more focused sensitivity experiments consisting of shorter four-month runs are discussed. This includes two runs forced by the two datasets ("HS" and "IS" for "HRRR Short" and "Interp Short", respectively) started from May 15 using an initial condition from a separate 2014 free model run ("FR14"). Comparison of the HS run

Table 1
Experimental setup for long and short runs with regular (HRRR and Interp) forcing datasets (NOAA = spun up by NOAA GLERL, FR14 = 2014 free run).

Runs using regular forcing			
Name	Period	Initialization	Forcing
HL	1/1/17–12/31/17	NOAA	HRRR
IL	1/1/17–12/31/17	NOAA	Interp
HS	5/15/17–9/15/17	FR14	HRRR
IS	5/15/17–9/15/17	FR14	Interp

Table 2
Experimental setup for five short runs with HRRR-Interp hybrid forcing datasets ($\mathbf{v}_w$ = wind velocity, T_a = air temperature, C = cloud fraction, T_d = dew point temperature, FR14 = 2014 free run, FR14S = 2014 free run plus spin-up). All runs use a lateral open boundary condition from NOAA GLERL.

Runs using hybrid forcing						
Name	Period	Initialization	$\mathbf{v}_w$	T_a	C	T_d
HF1	5/15/17–9/15/17	FR14S	HRRR	HRRR	HRRR	HRRR
HF2	5/15/17–9/15/17	FR14S	Interp	HRRR	HRRR	HRRR
HF3	5/15/17–9/15/17	FR14S	Interp	Interp	HRRR	HRRR
HF4	5/15/17–9/15/17	FR14S	Interp	Interp	Interp	HRRR
HF5	5/15/17–9/15/17	FR14S	Interp	Interp	Interp	Interp

to the HL run shows how quickly each of the model variables T, $\mathbf{v}$, ζ spins up to the HRRR forcing given different initial conditions, while comparison of both HS and IS runs to the HL run allows for a quantification of the relative importance of initialization versus forcing effects as the run progresses. A further series of short runs is designed to gauge the impact of each individual forcing variable (Table 2). This is done by forcing the model using a series of hybrids of the HRRR and Interp forcing datasets (runs "HF1" through "HF5", for "Hybrid Forcing"), beginning with the HRRR dataset itself (HF1), then switching one forcing variable at a time to the Interp version until all variables are from the Interp dataset (HF5). The May 15 initialization of these runs ("FR14S") uses a one-month ensemble-based spin-up whose Apr 15 initialization and forcing are based on the 2014 free run and the HRRR forcing, respectively (see our forthcoming DA-focused paper for details). The order in which the variables are switched ($\mathbf{v}_w$, T_a, C, T_d) is somewhat arbitrary, corresponding roughly with an expectation of decreasing order of importance. Of course, this setup does not exhaust all possible combinations of HRRR and Interp variables, but it at least provides a framework for examining the effect of each variable in isolation.

The only other differences between runs are that the long runs use double-precision floating-point arithmetic and hourly output (for

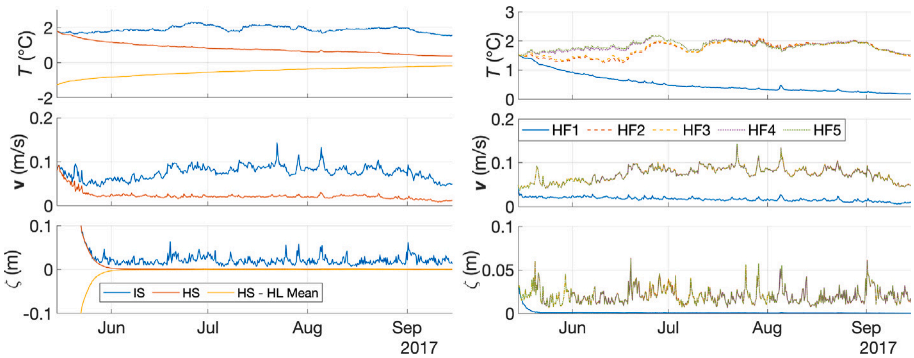


Fig. 2. Spatial-RMS differences from long HRRR (HL) run for short HRRR (HS) and short Interp (IS) runs with perturbed initialization (left), and hybrid-forced runs HF1 through HF5 (right; see Table 2). The mean of the HS-HL difference is also plotted for T and ζ. The larger RMSD for the IS run compared to HS shows that surface forcing differences eventually become more important than initialization differences for all variables, albeit at different rates for different variables. The HF1–HF5 plots show that wind velocity is the only surface forcing variable having an appreciable effect on $\mathbf{v}$ and ζ, while the cloud fraction has a smaller, transient effect on T as well.

maximal accuracy and resolution of transient events), while the short runs use single precision with output every six hours (to conserve resources and to align with the DA window for the runs presented in the authors' forthcoming DA-focused paper). All other aspects are identical across runs, including the lateral open boundary conditions, consisting of hourly water temperatures at the two inflow nodes and water levels every six minutes at the seven inflow/outflow nodes, based on data from NOAA/NOS gauges near Gibraltar, MI and Buffalo, NY (including an offset of 0.6 m for Buffalo due to the historically-based elevation difference between the gauge site and the model boundary location).

3. Results and discussion

3.1. Sensitivity of model and forcing variables

As a first step in understanding the sensitivity of the model to its surface forcing, results are presented here for the four-month model sensitivity experiments described in Section 2.3 and Tables 1 and 2. For the HS and IS runs, time series of the spatial root-mean-square difference (RMSD) from the HL run are plotted to determine (i) how long each variable in the HS run takes to spin up to a state close to the HL state, and (ii) how the magnitude of the HS vs. HL differences (due only to initialization) compares to those of the IS vs. HL differences (due to both initialization and forcing).

These time series are shown on the left side of Fig. 2. The HS run shows short spin-up periods of 1–2 weeks for $\mathbf{v}$ and 3–4 weeks for ζ. The longer response time of ζ compared to $\mathbf{v}$ in HS is due to adjustment of the lake-mean value, which is a response to the lateral, rather than surface, open boundary conditions. The adjustment to T also largely consists of an adjustment to the lake-mean value, but on a much longer time scale (more than four months). Across all variables, differences due to forcing are found to dominate those due to initialization after the spin-up period, with the RMSD eventually at least twice as great for the IS run as for the HS run across all variables. This is particularly true for ζ, where the RMSD for the HS run becomes negligible compared to the IS run after the spin-up period (during which the mean level is adjusted). This shows that ζ is the variable most sensitive to the winds, as might be expected, since the mean value is controlled by the inflow/outflow boundary conditions (unchanged across runs here), and the remaining variability is generally due to water pileup (time scales <1 day) and release through seiches (time scales up to a few days). For $\mathbf{v}$, the dominance of forcing differences over initialization differences also makes sense given the wind-driven circulation, but the RMSD of the HS run is no longer negligible, indicating that some memory of initial circulation differences is retained over the following months, though the HS circulation pattern (not shown) still appears very similar to that of HL. For T, the long spin-up time is likely related to the strong summer stratification, resulting in only slow penetration of the surface fluxes to the hypolimnion via turbulent diffusion and large-scale vertical mixing.

For the hybrid-forcing runs (HF1–HF5 in Table 2), the RMSD across the two runs is also plotted for T, $\mathbf{v}$ and ζ (as in the previous experiment), in order to gauge which forcing variable switches have the greatest effect on the model (right side of Fig. 2). As might be expected, $\mathbf{v}$ and ζ are negligibly affected by differences in any surface forcing variable other than wind. By contrast, T is significantly affected by changes in cloud cover over the first two months or so, with greater cloud cover in the Interp dataset (as shown in Section 3.2) causing colder T. Over time, however, these differences become dwarfed by those due to the different wind fields, indicating that the wind is again the most impactful forcing variable in the long run.

3.2. Comparison of HRRR and interpolated surface forcing

The HRRR and Interp June-July-August means for all surface forcing variables are compared in Fig. 3. For brevity, the discussion in this section uses the shorthand "WB" for western basin, "CB" for central basin, and "EB" for eastern basin (as defined in Fig. 1). The surface winds ($\mathbf{v}_w$) display broad similarities across the two datasets, with both showing a predominantly southwesterly direction aligned closely with the primary axis of the lake, with speeds generally increasing to the northeast (as wind fetch increases). However, significant local differences appear, particularly near the shoreline, where the HRRR winds show greater across-shore gradients and a more landward direction than the Interp winds. The largest differences occur along the north shore of the EB, where the more westerly Interp winds blow nearly parallel to the shore, leading to visibly greater wind convergence in the eastern part of the basin. The Interp winds are also stronger along most of the south shore, by as much as 1 m/s in parts of the CB. Finally, the Interp winds contain a distortion around Long Point which is not observed in the HRRR winds, consisting of a weakening by about 1 m/s and a strong turn toward the north. This is likely related to the relative isolation and unique water-surrounded nature of the Long Point observing station within the ground observing network used for interpolation.

A summary of the empirical orthogonal function (EOF) decomposition of the two wind fields is shown in Fig. 4. An EOF decomposition produces a series of orthonormal spatial modes (representing the principal deviation directions of the field from its time-mean value), a corresponding series of orthonormal temporal modes (giving the deviation coefficients over time), and a series of singular values (whose normalized squares give the explained variance of the field for each mode). The HRRR spatiotemporal modes are presented in order of decreasing strength, while the Interp modes are slightly shuffled so

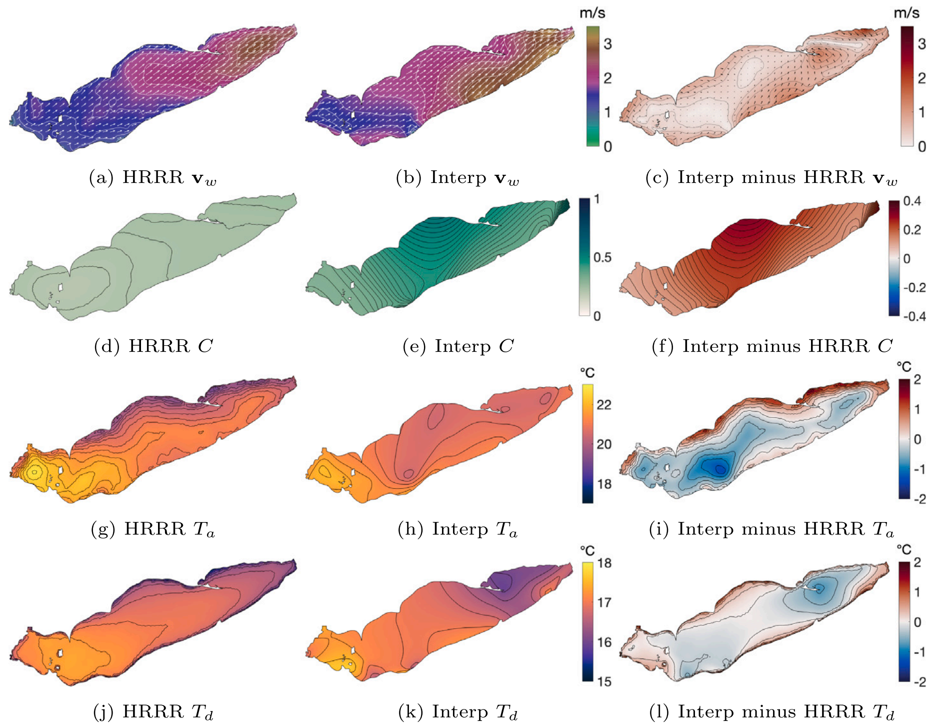


Fig. 3. HRRR vs. Interp mean surface meteorology over Jun-Jul-Aug, showing wind velocity ($\mathbf{v}_w$), cloud fraction (C), air temperature (T_a) and dew point temperature (T_d). Contour spacing is 20 cm/s for $\mathbf{v}_w$, 0.01 for C, and 0.25 °C for T_a and T_d.

that each HRRR mode is paired with whichever Interp mode is best-correlated (spatially) from the remaining ones (up to the 30th). For both datasets, most of the explained variance (86%–87%) comes from the first two modes, which are nearly uniform in space, and therefore control the general wind direction at any point in time, which explains their relative smoothness in time. The HRRR and Interp versions of these modes are closely correlated in both space and time, with the notable exceptions of the aforementioned Interp distortions in the EB (around Long Point and along the north shore). The remaining modes are more volatile in time, with their spatial components generally organized around a small number of elliptic/hyperbolic fixed points and stable/unstable manifolds. About 48%–52% of the remaining variance is explained by the third and fourth modes, which control opposing motions in the WB and EB. The fifth mode shows the diurnal cycle of the land/sea breeze, which for the Interp dataset, interestingly, is more pronounced along the south shore, where the majority of observing stations lie. The correlations of both spatial and temporal modes follow a downward trend, with temporal correlations falling below 0.5 for all modes after the first six, although a few spatial modes are quite well-correlated beyond this point.

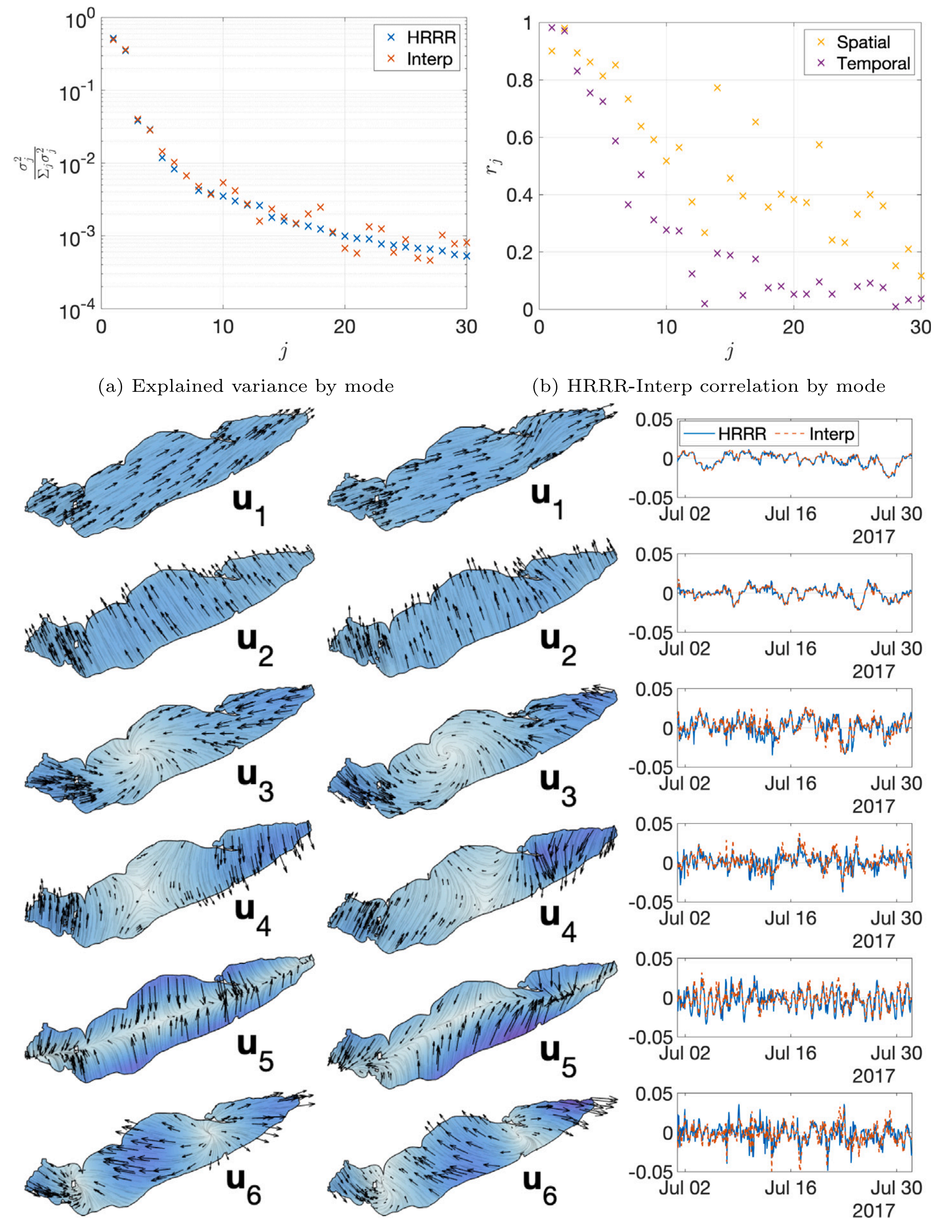
Returning to Fig. 3, among the non-wind forcing variables, the most pronounced difference between the HRRR and Interp means is the greater Interp cloud fraction (C) over the whole lake, especially over the CB, where it is as much as 30% greater on the north shore, representing a disparity in solar heating which likely drives the model T differences discussed in Section 3.1. As with the wind field, the Interp T_d shows a local distortion around Long Point, and the air and dew point temperature fields (T_a and T_d) display greater across-shore gradients in the HRRR dataset, with T_a along the north shore more than 5 °C lower than at the lake midline in some places. Despite these spatial variations, the HRRR-Interp difference for both T_a and T_d is less than 2 °C everywhere.

3.3. Model run using HRRR forcing

In this section the year-long 2017 model run using the HRRR forcing (the “HL” run) is investigated in detail, as a basis for comparison to the model run using the Interp forcing (the “IL” run) in the next section. Slowly-evolving flow features are discussed first, followed by transient features.

Fig. 5 shows monthly means of various fields of interest for June, July and August. The depth-averaged $\mathbf{v}$ field (Figs. 5(a)–5(c)) shows the development of an interesting circulation pattern consisting of two strong gyre currents (5–10 cm/s) in July and persisting into August: an oblong, bulbous anticyclonic (clockwise) gyre covering much of the CB, and a cyclonic (counterclockwise) gyre covering most of the EB. This cyclonic circulation in the EB is similar to several previous modeling and observational studies, where baroclinic mechanisms have been suggested (Beletsky et al., 2013), while the predominant anticyclonic circulation in the CB accords with some modeling and observational studies (Beletsky et al., 2013, 2012, 1999), but differs notably from a 15-year climatology produced using FVCOM (Bai et al., 2013), which used the lower-resolution North American Regional Reanalysis forcing and no inflow boundary condition. This greater variability of summer currents in the CB has generally been understood as an effect of its greater wind-dependence, due to its relatively flat, shallow nature (average depth ~18.3 m). Prior to this gyre development, currents are generally weak in June (0–2 cm/s), except near the shorelines, where in the relatively flat CB, they are northeastward and accompanied by a relatively simple overturning circulation consisting of north shore upwelling and south shore downwelling (Fig. 5(d)). As described in Rowe et al. 2019, this is in accordance with a southeastward Ekman transport due to the southwesterly mean winds (Fig. 3(a)). This pattern of Ekman transport generally continues in the deeper EB, but vertical currents are stronger and more complex due to the greater depth and steeper bathymetry (Fig. 1). In July and August, vertical mixing in the CB becomes more complex (Figs. 5(e), 5(f)), while the complex vertical mixing in the EB intensifies slightly, particularly in July.

The structure of the two gyres is reflected not only in the $\mathbf{v}$ field, but also in the T and ζ fields. Their period of maximal strength in July gives mean ζ differences of about 1–3 cm between the inside and outside of each gyre, with a raised surface inside the anticyclonic gyre, and lowered surface inside the cyclonic gyre (Figs. 5(g)–5(i)).



(a) Explained variance by mode (b) HRRR-Interp correlation by mode

(c) HRRR spatial modes 1–6 (d) Interp spatial modes 1–6 (e) Temporal modes (July)

Fig. 4. Empirical orthogonal function (EOF) decomposition for HRRR vs. Interp wind fields for all of 2017, showing explained variance and HRRR-Interp correlation (spatial and temporal) by mode, and the leading six spatial and temporal modes. Each HRRR mode (starting with the first) is paired with the spatially best-correlated of the remaining Interp modes. For the spatial plots, arrow density is insignificant, but streamlines and shading by magnitude are included.

This relationship is in accordance with geostrophic flow (i.e., a balance between the Coriolis force and the pressure gradient force) near the surface. The near-surface pressure gradient force is generally opposite the ζ gradient, while at depth the T and S distribution come into play because of the dependence of density on these variables (Li et al., 2008). The principle of geostrophy has been invoked many times to explain flows in the Great Lakes, as they are large enough for the Coriolis force to affect the large-scale flow.

For the T field, the monthly means were used to estimate thermocline depth as in Beletsky et al. (2012), using least-squares minimization on a three-layer piecewise-linear model (epilimnion, metalimnion, hypolimnion) of the T profile. The results (Figs. 5(j)–5(l)) show a deepening of the thermocline within the anticyclonic gyre, in a pattern corresponding closely to the raising of the surface height, by more than 2 m for every 1-cm change in ζ. Similarly, a shallower thermocline is observed in the cyclonic gyre, where ζ is lowered. These results accord with previous ones linking a bowl-shaped (dome-shaped) thermocline to anticyclonic (cyclonic) circulation and surface winds (respectively), in both observations (Beletsky et al., 2012) and models (Beletsky et al., 2013), wherein Ekman pumping/suction is posited as the primary mechanism. This explanation is further supported here by the pattern of increased downwelling seen within the CB in July (Fig. 5(e)), which corresponds closely with the CB anticyclonic gyre. It is also worth noting that the downward sloping of the CB thermocline from north to south shore in June accords with the aforementioned overturning circulation there. Finally, temperature profiles taken through the gyre centers (Figs. 5(m)–5(o)) offer further illustration of these bowl/dome thermocline shapes, as the thermocline deepens through the summer, reaching maximum strength in July.

The vertical structure of the gyre currents is further examined in Fig. 6. The surface currents are weaker in the southern half of the CB gyre, and in the northern half of the EB gyre. As the transects show, this

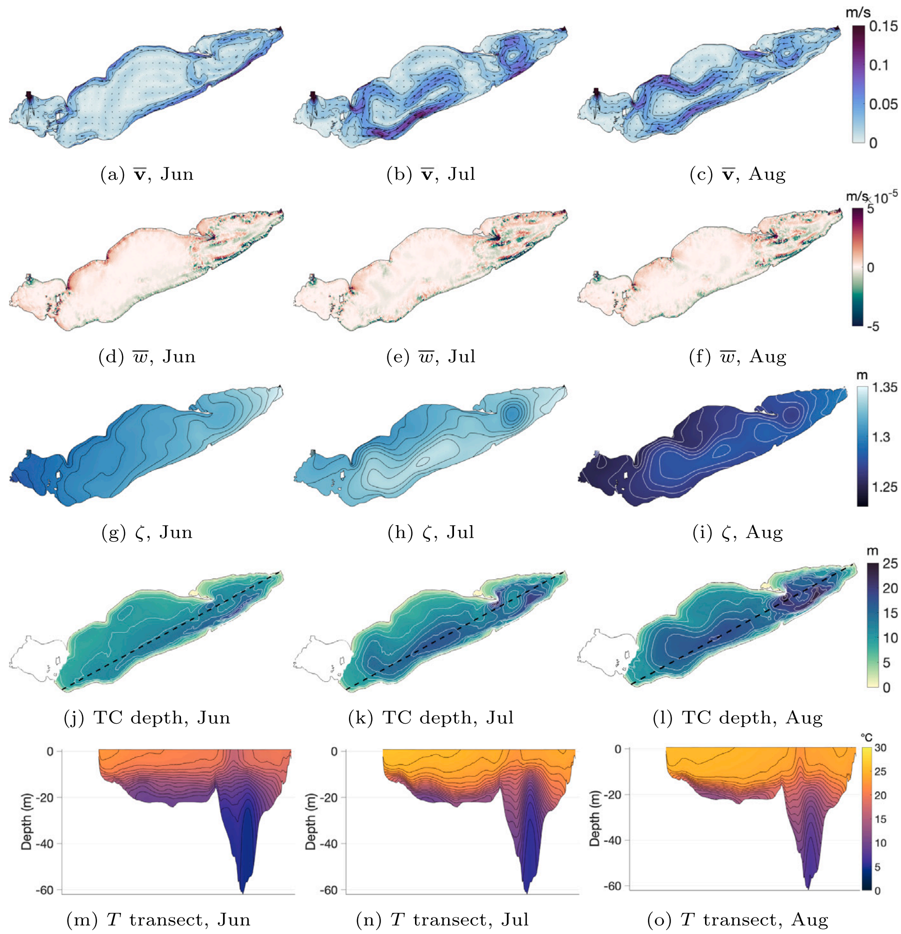


Fig. 5. Long HRRR (HL) run Jun, Jul, Aug monthly means for depth-mean horizontal and vertical velocities ($\overline{\mathbf{v}}$ and $\overline{w}$), free surface height (ζ), thermocline (TC) depth, and a transect of temperature (T) through the dashed line (contours: 2 cm/s, none, 0.5 cm, 2 m, and 1 °C, respectively). Strong gyre currents occur in the central and eastern basins in July and August, accompanied by corresponding changes in surface height and thermocline depth, consistent with geostrophic balance and Ekman pumping.

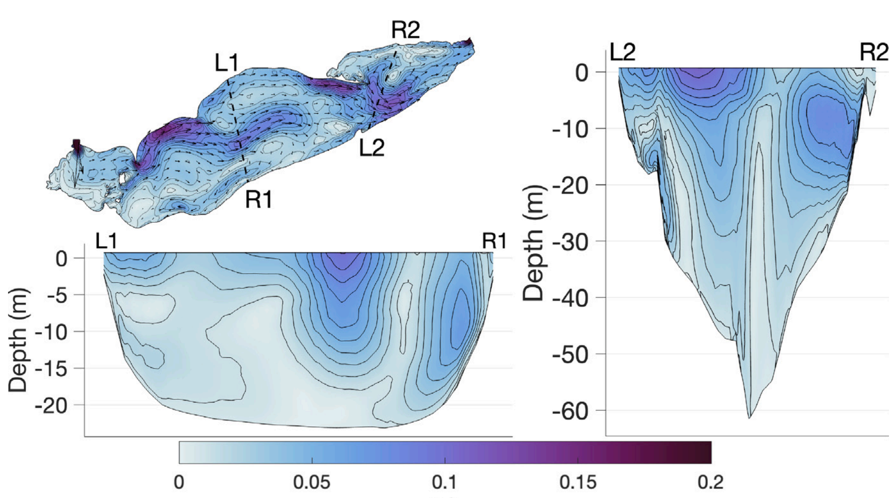


Fig. 6. Long HRRR (HL) run time-mean horizontal velocity $\mathbf{v}$ over Jun-Jul-Aug (with 1 cm/s contours), surface layer (top left) and two vertical transects. Both CB and EB gyres are tilted, but in somewhat opposing directions.

reflects an overall tilt to each gyre. For the CB gyre, the downward tilt from north to south is in accordance with the thermocline tilt and the overturning circulation there. The opposite tilt to the EB gyre is less-easily explained, given its opposition to the thermocline tilt direction and the complexity of vertical mixing in the EB.

Transient, oscillatory features in the HL run are revealed via spectral analysis of hourly time series for ζ and top-layer $\mathbf{v}$ during Jun-Jul-Aug, at points near the centers of each of the three basins (Fig. 7). This analysis produces power spectra for ζ, and clockwise and counter-clockwise rotary spectra for $\mathbf{v}$, using Welch's method with 17 windows

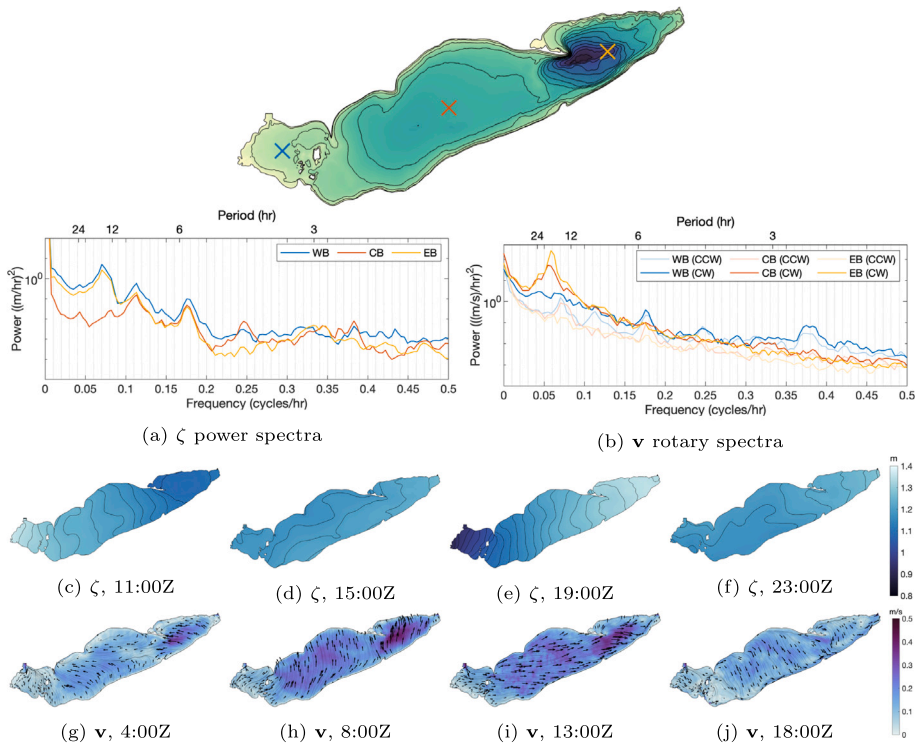


Fig. 7. Long HRRR (HL) run power spectra for free surface height ζ (a) and clockwise (CW) and counterclockwise (CCW) rotary spectra for top-layer horizontal velocity **v** (b) during Jun-Jul-Aug at three sites near centers of western basin (WB), central basin (CB) and eastern basin (EB), showing near-inertial oscillations and seiches. (c) - (f): ζ on September 5th (with 2-cm contours), showing seiche with slight counterclockwise component. (g) - (j): Top-layer **v** on June 4th (with streamlines), showing near-inertial oscillations in CB and EB.

of 256 hourly values and a Hann window function with 50% overlap across windows. Two well-known oscillatory phenomena are apparent as peaks in the data, namely seiches and near-inertial oscillations. In the rotary spectra, the clockwise component dominates across all basins, and the predominant feature is the peak with 17–18h period in the CB and especially EB, indicating the presence of near-inertial oscillations. In the ζ spectra, the primary seiche mode is seen as a peak in the WB and EB (where its amplitudes are highest) with a period of 14.2 h. Animations show that these seiches tend to have a slight counterclockwise component (e.g. Figs. 7(c)–7(f)), which likely accounts for the corresponding peaks in counterclockwise **v** spectra in the WB and CB. Additional seiche modes are seen across all basins at periods of about 8.8 h and 5.7 h, each accompanied by increased rotary activity in at least one basin. These near-inertial oscillation and seiche periods correspond closely to those found in previous observational studies (Boyce and Chiocchio, 1987; Saylor and Miller, 1987; Rao et al., 2008).

3.4. Comparison of model runs with different surface forcing

In Fig. 8, the HRRR-forced run illustrated in the previous section (referred to here as the "HL" run) is compared to the corresponding run using the Interp forcing (referred to as "IL"). For simplicity, the mean fields are shown for the month of July, when the HL run two-gyre circulation structure is most well-defined. The figure shows major differences in the CB and EB circulation across the two runs (Figs. 8(a)–8(c)). In the IL run, the HL gyre currents are relatively well-matched in the western part of the CB, but the strong southeastward current emerging from Rondeau Bay on the north shore weakens and continues to the south shore, rather than turning eastward to extend the loop as in the HL run. In the EB, the IL run circulation more closely resembles a two-cell structure, with an oblong anticyclonic cell in the north and a more circular cyclonic cell in the south, similar to the EB gyre in the HL run, but smaller and centered further to the south. The formation of an anticyclonic cell is likely closely linked to the difference in surface winds in the northern part of the EB (Figs. 3(a)–3(c)), with the stronger, more westerly nearshore Interp winds likely driving the more eastward nearshore IL currents. While the differences in EB vertical motions are quite complex, the westward current separating the two EB cells in the IL run is generally associated with greater downwelling, particularly in the northeast corner of the basin, where it begins (Figs. 8(d)–8(f)). This downwelling is likely induced by the greater convergence in the Interp surface winds in that region. As with the HL run, the IL run ζ appears largely in geostrophic balance with the currents, with gradients sloping upward from left to right relative to the current direction (Figs. 8(g)–8(i)). In particular, in the EB, a bulge and depression corresponding to the two circulation cells results in a ζ difference exceeding 2 cm from the HL run, while in the CB, the lack of strong gyre currents in the central part of the lake results in a ζ difference exceeding 1.5 cm from the HL run. For the thermocline depth (Figs. 8(j)–8(l)), the HL-IL differences induced by the circulation differences follow a similar spatial pattern, with the thermocline as much as 14 m deeper for IL than for HL in the EB near Long Point, and a more modest but still quite substantial 4 m deeper in the shallower CB, where the thermocline depth is an important determining factor for hypoxia development. Downward motions largely prevail within the respective (anticyclonic) CB gyres (Figs. 8(d)–8(f)), pointing again to the role of Ekman pumping.

In addition to these differences in the time-mean fields, the HL and IL runs also exhibit differences in transient, oscillatory phenomena. Fig. 9 shows power spectra (for ζ) and rotary spectra (for top-layer **v**) at sites near the basin centers for both the HL and IL runs, as well as their difference. The spectra of the IL run are similar to those of the HL run, indicating the presence of seiches and near-inertial oscillations with similar periods, with the most noticeable difference being the reduced power of some of the higher-frequency seiche modes. More interestingly, the IL–HL differences exhibit spectral peaks with

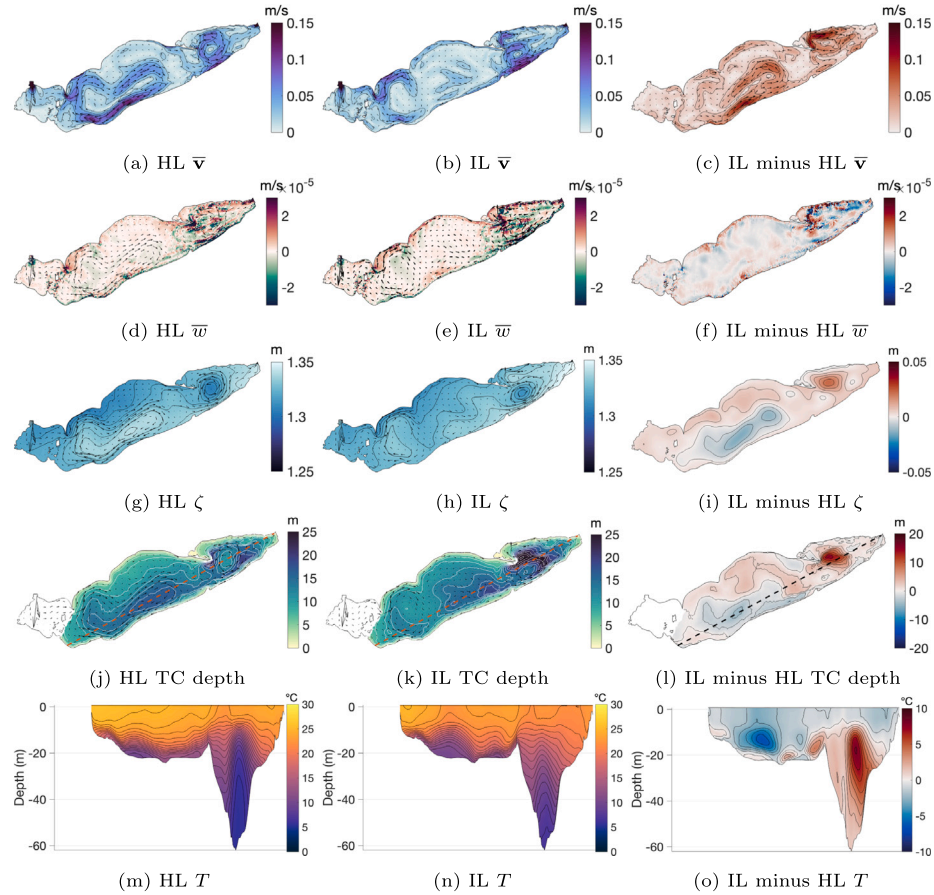


Fig. 8. Long HRRR (HL) versus long Interp (IL) runs: July-mean vertically-averaged horizontal and vertical velocities ($\overline{\mathbf{v}}$ and $\overline{w}$) and free surface height ζ, as well as thermocline (TC) depth calculated from July-mean T (with contours of 2 cm/s, 0.01 mm/s, 0.5 cm, 2 m, and 1 °C, respectively). The $\overline{\mathbf{v}}$ field is superimposed in the bottom two rows to show its close correspondence to ζ and TC depth, in accordance with geostrophy and Ekman pumping/suction.

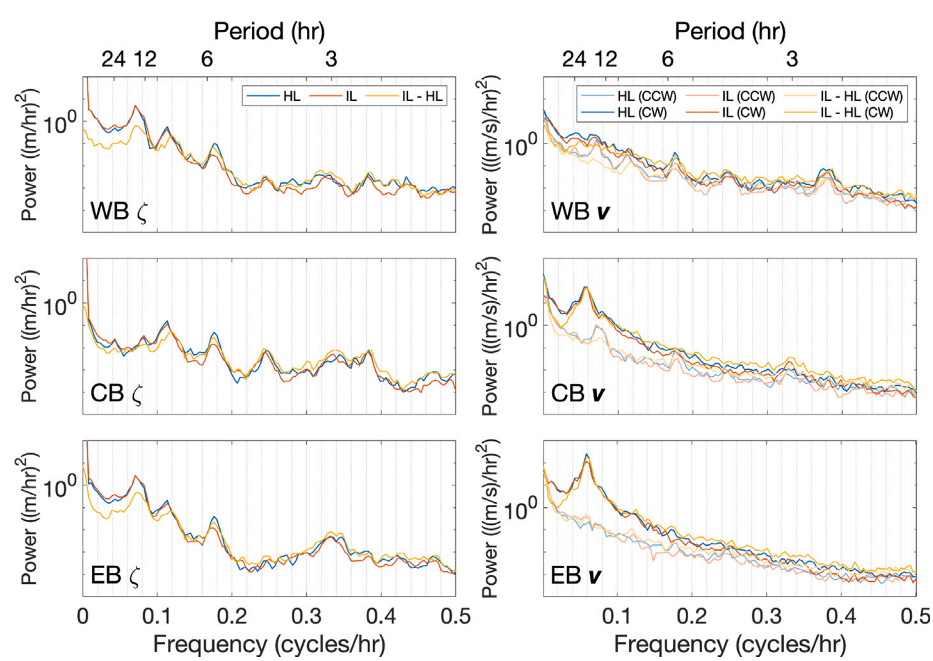
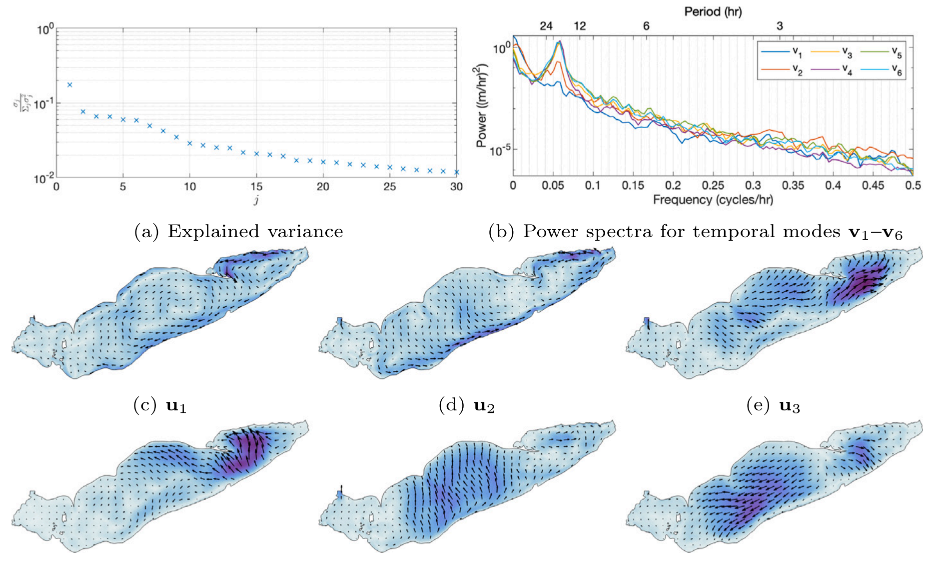


Fig. 9. Power spectra for free surface height ζ and rotary spectra for top-layer horizontal velocity $\mathbf{v}$ for long HRRR (HL) and long Interp (IL) runs as well as their difference at sites near the three basin centers during Jun-Jul-Aug. Overall, the HL run, the IL run and their difference all exhibit similar peaks, while the differences show lower power at low frequencies for ζ in the WB and EB and for $\mathbf{v}$ in the CB and EB, and somewhat higher power at high frequencies for $\mathbf{v}$ in the CB and EB.



(a) Explained variance (b) Power spectra for temporal modes $\mathbf{v}_1$–$\mathbf{v}_6$

(c) $\mathbf{u}_1$ (d) $\mathbf{u}_2$ (e) $\mathbf{u}_3$

(f) $\mathbf{u}_4$ (g) $\mathbf{u}_5$ (h) $\mathbf{u}_6$

Fig. 10. Empirical orthogonal function (EOF) decomposition of difference in horizontal velocity $\mathbf{v}$ between long HRRR (HL) and long Interp (IL) runs during Jun-Jul-Aug: Explained variance, power spectra of leading temporal modes ($\mathbf{v}_1$–$\mathbf{v}_6$), and top layer of leading spatial modes ($\mathbf{u}_1$–$\mathbf{u}_6$). Modes 1–2 control alongshore currents on the north shore of the EB and the south shore of the CB; modes 3–4 and 5–6 control near-inertial oscillations in the EB and CB, respectively.

magnitudes similar to those of the runs themselves. This suggests that, although seiches and near-inertial oscillations are present in similar quantities in the two runs, the details of these events differ substantially across the two runs. Such details might include their precise amplitude, phase/timing, or location, and the corresponding differences across runs are consistent with the fast response time of $\mathbf{v}$ and especially ζ to the surface winds (as shown in Section 3.1).

A couple of other themes emerge from Fig. 9. One of these is that the low-frequency end of the spectrum exhibits less power in the IL–HL difference than in the runs themselves. For $\mathbf{v}$, this is mostly restricted to frequencies less than 0.05 cycles/hr (i.e. periods longer than 20 h), but for ζ in the WB and EB, it extends all the way to about 0.12 cycles/hr (8-hour period), including the two dominant seiche modes. Conversely, for CB and EB $\mathbf{v}$ at least, the high-frequency end of the spectrum is more powerful in the IL–HL differences than in the runs themselves. This suggests that the slowly-evolving mean circulation pattern, and associated changes to the thermocline and surface height, may be in some sense less sensitive to the choice of surface forcing than transient, oscillatory phenomena (though still quite sensitive, as seen in the previous analysis).

While the preceding spectral analysis of the IL–HL differences in $\mathbf{v}$ and ζ focuses on specific locations within the lake, Figs. 10 and 11 take the whole lake into account by presenting an EOF for the $\mathbf{v}$ and ζ differences in space and time, with the temporal modes presented through their power spectra. Note that in an EOF, the temporal mean is subtracted off before a singular value decomposition is performed, so that the information presented here does not overlap with the mean differences presented in Fig. 8. In the $\mathbf{v}$-difference EOF (Fig. 10), the largely orthogonal surface directions of spatial mode pairs 3–4 and 5–6, combined with the temporal modes' strong spectral peaks around a period of 17–18h and approximate 90° phase difference (not pictured), suggest that these mode pairs predominantly represent differences in near-inertial oscillations in the EB and CB, respectively. Modes 1 and 2 are generally associated with the narrow, swift coastal currents along the CB south shore and the EB north shore (where the steepest bathymetry occurs). Mode 1 is relatively lacking in spectral peaks, while mode 2 shows a moderate peak at the near-inertial oscillation period. While mode 1 is the dominant mode by far, accounting for 17.4% of the explained variance, the 25.0% accounted for by modes 3–6 shows that near-inertial oscillations form a substantial component of the IL–HL run differences in $\mathbf{v}$.

In the ζ-difference EOF (Fig. 11), clear spectral peaks are seen in temporal modes 1–6, indicating oscillatory behavior in each mode. In mode 1, the spatial-mode pattern and temporal-mode peak near a 14-h period correspond to those of the primary (lowest-frequency) seiche mode. In modes 2–6, the spectral peaks at periods of approximately 8.4 h, 5.7 h, 4.1 h, 3.0 h and 2.6 h correspond largely to those of higher seiche modes, with modes 4–6 in particular showing unique highest peaks and typical standing-wave patterns. With modes 1–6 accounting for 87.3% of the explained variance, this suggests that seiche effects form an even larger component of the ζ differences between IL and HL runs than near-inertial oscillations do for the $\mathbf{v}$ differences.

Fig. 12 uses time series of u and ζ at three locations each to illustrate how a substantial portion of the differences between the HL and IL runs can result from differences in the phase and amplitude of oscillations in these variables. For example, near-inertial oscillations (seen in the u plots) are nearly in phase in the EB on July 7th and 21st, but fall somewhat out of phase in between those dates. Differences in u at the CB location show a similar phase pattern, compounded by the previously-discussed systematic differences in the mean fields. In the ζ plots, the IL run exhibits stronger seiche amplitudes near the ends of the lake from July 20th–31st, while July 17th–19th shows somewhat the opposite pattern, accompanied by slight phase differences.

Collectively, therefore, seiches and near-inertial oscillations comprise a major component of the model differences observed when using different forcing datasets. This fact has important implications for real model forecasting applications, since it suggests that inevitable errors in the model wind forcing can be expected to cause substantial errors in short-term forecasting of these phenomena. More generally, the large influence of these transient phenomena should be kept in mind whenever comparing two model runs with different forcing datasets, such as in the OSSE presented in the authors' forthcoming DA-focused paper, where both free runs and DA runs using the Interp forcing are compared against a nature run using the HRRR forcing.

4. Conclusions

This work investigates the sensitivity of a 2016 version of the Lake Erie FVCOM model used by LEOFS to the choice of atmospheric surface

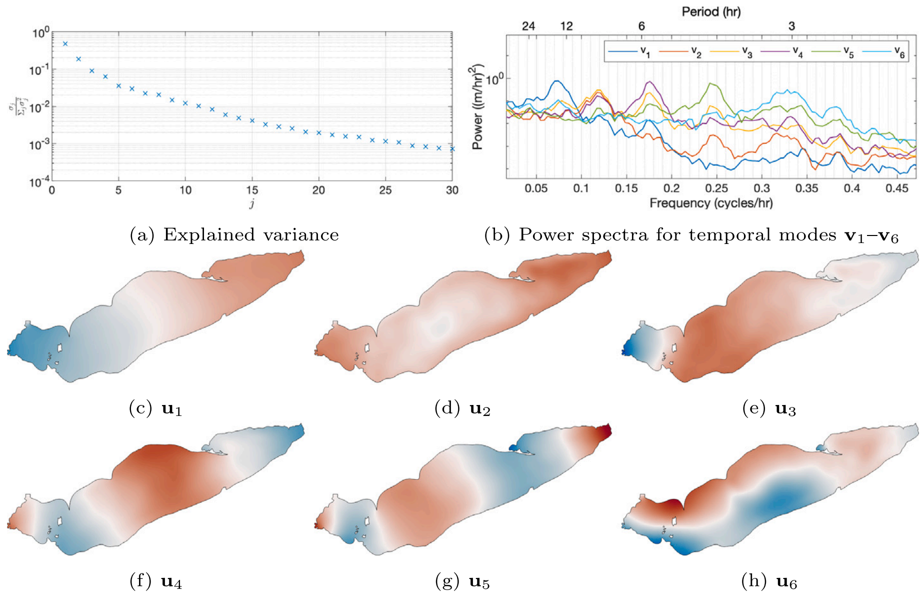


Fig. 11. Empirical orthogonal function (EOF) decomposition of difference in free surface height ζ between long HRRR (HL) and long Interp (IL) runs during Jun-Jul-Aug: Explained variance, power spectra of leading temporal modes ($\mathbf{v}_1$–$\mathbf{v}_6$), and leading spatial modes ($\mathbf{u}_1$–$\mathbf{u}_6$). Mode 1 is the dominant seiche mode, while modes 2–6 represent higher seiche modes.

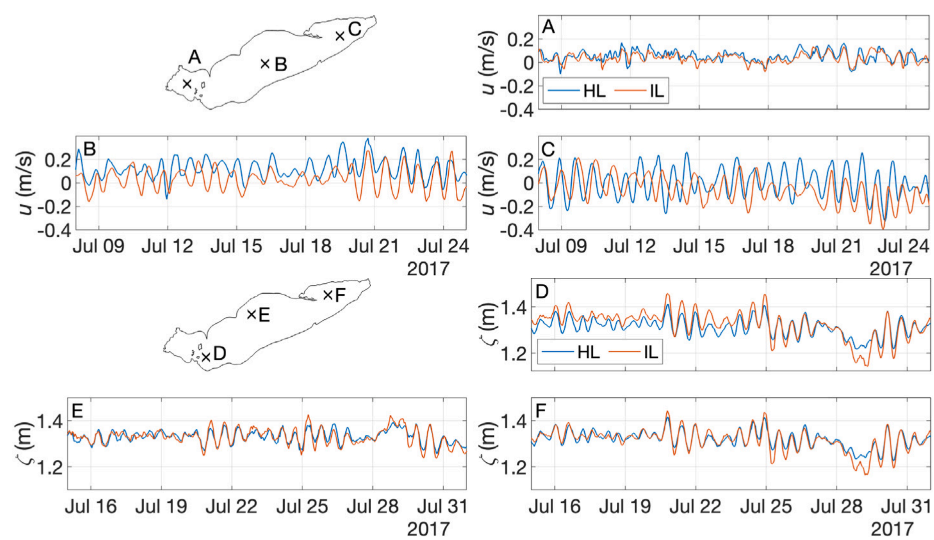


Fig. 12. Time series of HL vs. IL top-layer u (top) and ζ (bottom) at three locations each. Differences in both variables can be partly attributed to differences in the phase and amplitude of near-inertial oscillations (for u) and seiches (for ζ).

forcing dataset during the summer months, when two of the lake's most important potential hazards frequently occur, namely HABs and hypoxic bottom waters. Two different surface forcing datasets that have been used in operations are used to force the model for all of 2017: one coming from NOAA's HRRR model, and the other based on observations from ground-based stations interpolated to the model grid. These datasets are shown to have broadly similar characteristics, as seen in their mean fields as well as their spatiotemporal EOF decomposition, albeit with some local differences (notably coastal direction and magnitude, and convergence over the eastern basin), and greater fine-scale structure in the HRRR dataset.

Despite these broad similarities, running the model for a year with each forcing dataset yields horizontal and vertical summer circulation patterns that differ noticeably from each other (as first shown in Rowe et al. 2019), as strong gyre currents develop in the central and eastern basins in July and August, an important period for HAB and hypoxia onset and transport. In both runs, this circulation is comprised principally of a cyclonic gyre in the eastern basin, similar to previous studies, and an anticyclonic gyre in the central basin which differs somewhat from previous studies. These mean circulation differences are accompanied by corresponding differences in the shape of the surface (consistent with geostrophic balance) and the thermocline (consistent with Ekman pumping/suction). While these mean differences evolve relatively slowly, differences in velocity and especially free surface height also contain substantial components that vary on short (< 1 day) time scales. The spectral peaks in the frequency of these differences match those of two well-known oscillatory phenomena which regularly show up in observations, namely seiches and near-inertial oscillations. While both forcing datasets simulate these phenomena in comparable amounts, the spectral peaks in the model-run differences suggest that details such as phase and amplitude are somewhat different across the two runs. A series of four-month model sensitivity experiments demonstrates that the winds are the dominant surface forcing variable driving all of these differences in model trajectory, although cloud cover also has a smaller role to play in driving surface temperature in particular (via solar heating), and that these wind-driven differences tend to dominate over initialization differences, in a matter of weeks for $\mathbf{v}$ and ζ, and months for the three-dimensional T field.

While one obvious implication of this work is the importance of choosing the best possible wind field to drive any wind-sensitive forecast model, in practice wind forcing errors are inevitable due to incomplete knowledge of the true surface winds as they vary in space and time. The differences shown here between the two differently-forced model runs may be taken as some indication of the types and magnitudes of forecast errors that may result from wind forcing errors in a free-running forecast model such as the one currently used by LEOFS. In particular, the substantial differences in both the mean flow and transient features suggest a large degree of uncertainty in several aspects of the model forecast, such as the overall circulation pattern, thermocline depth and shape, and the precise phase, amplitude, timing, and location of near-inertial oscillations and seiches. This implies uncertainty in applications such as forecasting seiche-induced flooding, the onset and spread of HABs, the development and extent of hypoxic bottom waters, and the long-term distribution of plastic pollution.

The inevitability of such forecast errors points to the value of regularly adjusting the forecast toward recent observations, rather than simply using a free-running model for the forecast. This process, typically referred to as data assimilation (DA), has long been a staple of forecast systems of the global atmosphere and oceans, and underlies much of the increasing skill of weather forecasts over the last several decades. The Great Lakes are an excellent candidate for incorporation of DA into the nowcast/forecast system given the mature state of GLOFS and the abundance of both satellite and in-situ observations of the lakes. A forthcoming paper by the authors will present and evaluate an ensemble-based DA system for Lake Erie which focuses specifically on assimilating temperature observations from both the surface and subsurface.

CRediT authorship contribution statement

David Russell: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Matthew J. Hoffman:** Writing – review & editing, Supervision, Software, Resources, Methodology, Investigation, Conceptualization. **Kayo Ide:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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